

AI Large Model Image Usage Instructions

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Important Notes Before You Begin

Parameter Description

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The AI large model development tutorial for RDK X5 requires the use of our pre-configured large model image: `System_File\RDK_X5_AI_Pure.zip`

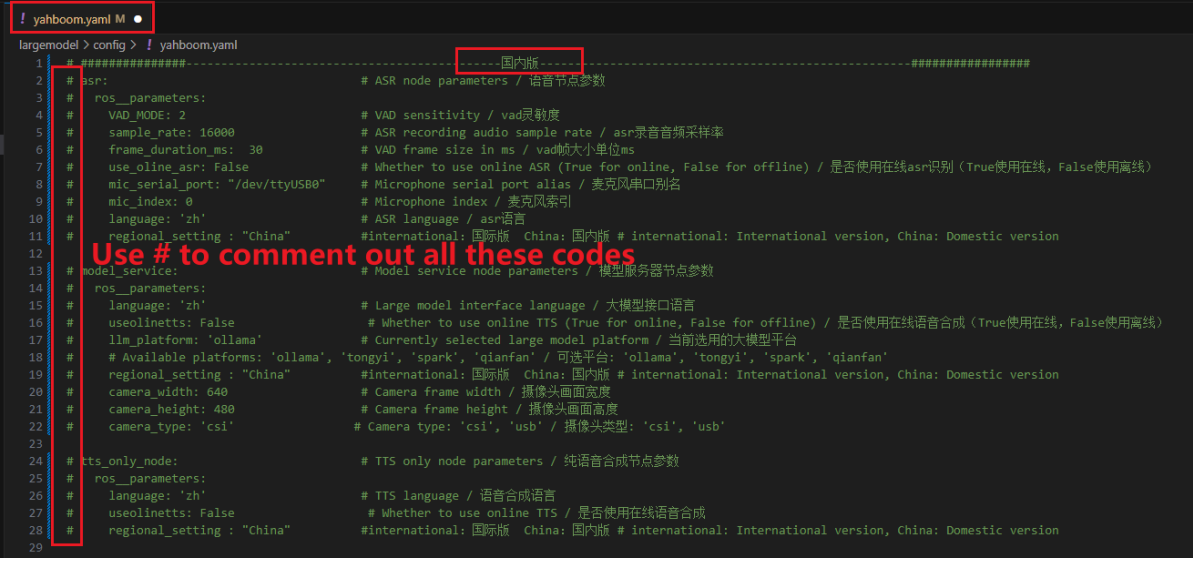
1. For users who have purchased Yahboom's RDK X5 mainboard, the TF card you receive is empty and has no image. You need to burn the image yourself. For detailed burning methods, please refer to the tutorial in the "1.Introduction to RDK X5 board/2.Write system file" section. The large language model image is only applicable to running the AI large model development chapter. If you want to run chapters other than the AI large model development, please burn the image in the "System_File\RDK_X5_20250227" folder.

2. Since running a large model with Ollama requires a lot of memory, motherboards with small memory can only use models with a small number of parameters, or even fail to run them. Sometimes when the running memory is at its limit, the effect will be poor. For a better experience with this function, please refer to the online large model chapter.

3. It is normal for the AI large model chapter to start slowly and lag when running.

4. For users outside of mainland China, please edit the following file:

`~/yahboom_ws/src/largemodel/config/yahboom.yaml`. In this file, comment out the line for the "domestic version" and uncomment the line for the "International Edition".



```
! yahboom.yaml M
largemodel > config > ! yahboom.yaml
1 # -----国内版-----
2 # asr: # ASR node parameters / 语音节点参数
3 # ros__parameters:
4 #   VAD_MODE: 2 # VAD sensitivity / vad灵敏度
5 #   sample_rate: 16000 # ASR recording audio sample rate / asr录音音频采样率
6 #   frame_duration_ms: 30 # VAD frame size in ms / vad帧大小单位ms
7 #   use_online_asr: False # Whether to use online ASR (True for online, False for offline) / 是否使用在线asr识别 (True使用在线, False使用离线)
8 #   mic_serial_port: "/dev/ttyUSB0" # Microphone serial port alias / 麦克风串口别名
9 #   mic_index: 0 # Microphone index / 麦克风索引
10 #   language: 'zh' # ASR language / asr语言
11 #   regional_setting: "China" #international: 国际版 China: 国内版 # international: International version, China: Domestic version
12 # model_service: # Model service node parameters / 模型服务器节点参数
13 # ros__parameters:
14 #   language: 'zh' # Large model interface language / 大模型接口语言
15 #   useonline tts: False # Whether to use online TTS (True for online, False for offline) / 是否使用在线语音合成 (True使用在线, False使用离线)
16 #   llm_platform: 'ollama' # Currently selected large model platform / 当前选用的大模型平台
17 #   # Available platforms: 'ollama', 'tongyi', 'spark', 'qianfan' / 可选平台: 'ollama', 'tongyi', 'spark', 'qianfan'
18 #   regional_setting: "China" #international: 国际版 China: 国内版 # international: International version, China: Domestic version
19 #   camera_width: 640 # Camera frame width / 摄像头画面宽度
20 #   camera_height: 480 # Camera frame height / 摄像头画面高度
21 #   camera_type: 'csi' # Camera type: 'csi', 'usb' / 摄像头类型: 'csi', 'usb'
22
23 # tts_only_node: # TTS only node parameters / 纯语音合成节点参数
24 # ros__parameters:
25 #   language: 'zh' # TTS language / 语音合成语言
26 #   useonline tts: False # Whether to use online TTS / 是否使用在线语音合成
27 #   regional_setting: "China" #international: 国际版 China: 国内版 # international: International version, China: Domestic version
28
29
```

Use # to comment out all these codes

```

30 #####-----International Edition-----#####
31 asr: # ASR node parameters / 语音节点参数
32   ros_parameters:
33     VAD_MODE: 2 # VAD sensitivity / vad灵敏度
34     sample_rate: 16000 # ASR recording audio sample rate / asr录音音频采样率
35     frame_duration_ms: 30 # VAD frame size in ms / vad帧大小单位ms
36     use_online_asr: False # Whether to use online ASR (True for online, False for offline) / 是否使用在线asr识别 (True使用在线, False使用离线)
37     mic_serial_port: "/dev/ttyUSB0" # Microphone serial port alias / 麦克风串口别名
38     mic_index: 0 # Microphone index / 麦克风索引
39     language: 'en' # ASR language / asr语言
40     regional_setting: "international" #international: 国际版 China: 国内版 # International: International version, China: Domestic version
41 Delete all previously existing #
42 model_service: # Model service node parameters / 模型服务器节点参数
43   ros_parameters:
44     language: 'en' # Large model interface language / 大模型接口语言
45     useonline tts: False # Whether to use online TTS (True for online, False for offline) / 是否使用在线语音合成 (True使用在线, False使用离线)
46     llm_platform: 'ollama' # Currently selected large model platform / 当前选用的大模型平台
47     # Available platforms: 'ollama', 'openrouter' / 可选平台: 'ollama', 'openrouter'
48     regional_setting: "international" #international: 国际版 China: 国内版 # International: International version, China: Domestic version
49     camera_width: 640 # Camera frame width / 摄像头画面宽度
50     camera_height: 480 # Camera frame height / 摄像头画面高度
51     camera_type: 'usb' # Camera type: 'csi', 'usb' / 摄像头类型: 'csi', 'usb'
52   tts_only_node: # TTS only node parameters / 纯语音合成节点参数
53     ros_parameters:
54       language: 'en' # TTS language / 语音合成语言
55       useonline tts: False # Whether to use online TTS / 是否使用在线语音合成
56       regional_setting: "international" #international: 国际版 China: 国内版 # International: International version, China: Domestic version
57
58
59
60

```

5.Since Tongyi Qianqian, Xunfei Starfire, and Baidu Qianfan AI big models need Chinese ID to register, international version users are only supported to use the openrouter online platform. In the yahboom.yaml file, when you want to use the online AI platform, be sure to set the llm_platform parameter to openrouter

System Information

Based on rdk_os_3.3.1-2025-8-8

Username: sunrise

User password: sunrise

System Environment

- Ollama
- Docker
- Open WebUI
- Offline large model storage path: /usr/share/ollama/.ollama/models

Due to the limitations of the SD (TF) card, we have not downloaded all the models in the tutorial to the system. We have deleted the large parameter version models in some tutorials. Users can delete and download the models themselves.

Parameter Description

RDK X5 (8G RAM): Runs models with 8B and below parameters

RDK X5 (4G RAM): Runs models with 3B and below parameters

The above conclusions are not completely accurate, but can be used as a reference!

ollama official website model running instructions: running a 7B parameter model requires at least 8G of available running memory, running a 13B parameter model requires at least 16G of available running memory, and running a 33B parameter model requires at least 32G of available running memory.